

Pranjal Gupta

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[GitHub](#) | [LinkedIn](#) | [LeetCode](#) | [Portfolio](#)

Education

University of Delhi – Faculty of Technology

2023 – 2027

B.Tech, Computer Science & Engineering (AI & ML)

CGPA: 7.60 / 10.0

Technical Skills

- **Languages:** Java, Python, JavaScript
- **Frontend:** React.js, HTML, CSS, Tailwind CSS
- **Backend:** Node.js, Express.js, Flask, FastAPI, REST APIs, JWT Authentication, Socket.IO
- **Databases:** MongoDB, MySQL, ChromaDB
- **AI/ML & GenAI:** PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases
- **Tools:** Git, GitHub, Docker, AWS (EC2, S3)
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming

Projects

Real-Time Multi-Chat Application

[GitHub](#)

- Built a full-stack real-time messaging platform using React, Node.js, Express, MongoDB, and Socket.IO supporting private conversations, persistent chat history, and live message delivery
- Implemented JWT-based authentication with access and refresh tokens, secure session management, room-based messaging architecture, and automatic socket reconnection handling
- Designed scalable REST APIs and MongoDB schemas for user management, chat creation, message storage, and conversation retrieval

Repository-Aware Coding Agent

[GitHub](#)

- Developed a CLI-based coding assistant using LangChain, Ollama, ChromaDB, and Retrieval-Augmented Generation (RAG) to answer questions over entire code repositories
- Implemented repository indexing, semantic code retrieval, file search, directory exploration, and code-editing tools for efficient codebase navigation
- Built a persistent vector database pipeline supporting large multi-file repositories through automated document ingestion and chunking

Multi-Agent System for Interview Preparation

[GitHub](#) | [Live](#)

- Built and deployed a multi-agent interview preparation platform using LangGraph, LangChain, FastAPI, Groq, Gemini, and ChromaDB, serving AI-powered question generation and answer synthesis workflows.
- Implemented Retrieval-Augmented Generation (RAG), vector search, embedding pipelines, and agent orchestration to provide context-aware interview preparation from user-provided documents.
- Developed scalable backend APIs and workflow automation, reducing manual preparation effort through automated content generation and personalized learning assistance.

Personal Portfolio with AI Assistant

[GitHub](#) | [Live](#)

- Built and deployed a personal portfolio website using React, FastAPI, Tailwind CSS, and modern responsive UI design principles
- Integrated a document-grounded AI assistant using LangChain, Gemini Embeddings and ChromaDB to enable contextual question answering about projects, skills and experience.

Achievements & Activities

- Volunteer, Indian DefSpace Symposium 2025 — Coordinated technical session logistics for a national-level symposium
- Training & Placement Committee, Faculty of Technology, University of Delhi